

# Module 4 — Lab: Descriptive Analytics

Conduct a complete exploratory data analysis on a multi-variable dataset for Hashemite Technical University and produce a findings report with statistical backing.

**Due:** Day 3 (Tuesday) end of day. Resubmissions are accepted through Saturday of the assignment week.

## Setup

- Accept the assignment:** [Accept the M4 Lab assignment](#)
- Clone your repo and open it in VS Code.
- The repo contains `data/student_performance.csv` — a dataset of ~2,000 student records from Hashemite Technical University.
- Install dependencies: `pip install -r requirements.txt`

No Docker or database setup needed — the data is in the CSV.

## The Dataset

Hashemite Technical University student academic records across departments:

| Column                          | Type  | Notes                                                                    |
|---------------------------------|-------|--------------------------------------------------------------------------|
| <code>student_id</code>         | str   | Unique identifier (e.g., <code>STU00001</code> )                         |
| <code>department</code>         | str   | Biology, Business, Computer Science, Engineering, Mathematics            |
| <code>semester</code>           | str   | Fall 2023, Spring 2024, Fall 2024                                        |
| <code>course_load</code>        | int   | Number of courses taken (3–7)                                            |
| <code>study_hours_weekly</code> | float | Self-reported weekly study hours                                         |
| <code>gpa</code>                | float | Semester GPA (0.0–4.0)                                                   |
| <code>attendance_pct</code>     | float | Percentage of classes attended                                           |
| <code>has_internship</code>     | str   | Currently doing an internship ( <code>"Yes"</code> / <code>"No"</code> ) |
| <code>commute_minutes</code>    | int   | Daily commute time; some missing                                         |
| <code>scholarship</code>        | str   | Athletic, Department, Merit, Need-based, None                            |

**Data characteristics to be aware of:**

- `commute_minutes` has ~9% missing values (181 out of 2,000 rows)
- GPA distribution is left-skewed (most students cluster 2.5–3.5 with a tail toward lower GPAs)
- There is a moderate positive correlation between `study_hours_weekly` and `gpa`
- Students with internships tend to have higher GPAs

Explore the data before diving into analysis: `df.shape` , `df.info()` , `df.describe()` , `df.head()` .

## Tasks

Complete the implementation of `eda_analysis.py` . The file contains function signatures with docstrings — you implement the logic.

### Task 1 — Data Inspection and Cleaning

Load the dataset. Report the following in `output/data_profile.txt` :

- Shape (rows, columns)
- Data types for each column
- Missing values per column (count and percentage)
- Your handling decision for each column with missing values (and the reasoning)

Handle missing values with documented reasoning. For example: if `commute_minutes` is MCAR and represents ~9% of rows, you might impute with the median or drop those rows. Justify your choices.

### Task 2 — Distribution Analysis

Create distribution plots for at least 3 numeric columns:

- Histograms with KDE** for continuous variables (e.g., `gpa` , `study_hours_weekly` , `attendance_pct` ) — use `sns.histplot(kde=True)`
- Box plot** comparing `gpa` across `department` — use `sns.boxplot()`
- Bar chart** for at least one categorical variable distribution (e.g., `scholarship` counts)

Save all plots to `output/` as PNG files.

### Task 3 — Correlation Analysis

- Compute the Pearson correlation matrix for all numeric columns
- Visualize as an annotated heatmap: `sns.heatmap(corr_matrix, annot=True, cmap='coolwarm')`
- Create scatter plots for the two most correlated variable pairs (excluding self-correlation)
- Document findings in your analysis: which variables are related, and what might explain the relationship?

Save all plots to `output/` .

### Task 4 — Hypothesis Testing

Test at least two hypotheses:

**Hypothesis 1:** “Students with internships have a higher GPA than students without internships.”

- Use an independent samples t-test: `scipy.stats.ttest_ind()`
- Report: t-statistic, p-value, and a plain-language interpretation
- Compute Cohen's d for effect size

**Hypothesis 2:** “Scholarship status is associated with department.”

- Use a chi-square test: `pd.crosstab()` then `scipy.stats.chi2_contingency()`
- Report: chi-square statistic, p-value, degrees of freedom, and a plain-language interpretation

Print all test results to the console and include them in your findings report.

### Task 5 — Findings Report

Write `FINDINGS.md` summarizing your analysis:

- Dataset description:** Shape, columns, notable data quality issues
- Key distribution findings:** Which variables are skewed? What do the box plots reveal about department differences?
- Notable correlations:** Which pairs are most correlated? What might explain the relationships? Include caveats (correlation is not causation).
- Hypothesis test results:** For each test — the hypothesis, test used, results (statistic + p-value), and plain-language interpretation. Was the result statistically significant? Is the effect practically meaningful (effect size)?
- Three actionable recommendations** for the university based on your findings. Each recommendation should reference specific data findings.

Include references to saved chart files (e.g., “See `output/gpa_by_department.png`”).

## Submit

- Create branch `lab-4-descriptive-analytics`
- Commit: `eda_analysis.py` , `FINDINGS.md` , and the `output/` directory (all generated files)
- Open a PR to `main`
- Your PR description must include:
  - The t-test result (t-statistic and p-value) for the internship/GPA hypothesis
  - One key finding from your analysis
- Paste your PR URL into TalentLMS → Module 4 → Lab 4 to submit this assignment.

## Challenge Extensions

These are optional extensions for learners who complete the base requirements. Each tier introduces concepts or techniques not covered in the base work.

### Tier 1 — Advanced Statistical Analysis

- Add an **ANOVA test**: “Does average GPA differ across the five departments?” Report and interpret the F-statistic and p-value. If significant, run post-hoc pairwise t-tests with **Bonferroni correction** to identify which departments differ.
- Create **violin plots** (not just box plots) for the department GPA comparison, showing distribution shape alongside summary statistics. Use `sns.violinplot()` .

### Tier 2 — Automated EDA Report Generator

- Build a reusable `eda_report.py` module that accepts **any DataFrame** and automatically generates:
  - Data profile (shape, types, missing values)
  - Distribution plots for all numeric columns
  - Correlation heatmap
  - Missing data visualization
  - Outlier summary (IQR method)
- Make it configurable: column selection, output directory, plot style.
- Write **tests** for the module: verify it handles DataFrames of varying shapes, column types, and missing data patterns.

### Tier 3 — Statistical Simulation and Power Analysis

- Implement a **bootstrap confidence interval** calculator: resample the data 10,000 times to compute 95% CIs for mean GPA by internship status. Compare with the parametric t-test confidence interval.
- Conduct a **power analysis**: given the observed effect size between internship/no-internship groups, what sample size is needed to detect the difference with 80% power at  $\alpha = 0.05$ ? Use `statsmodels.stats.power.TTestIndPower` .
- Create a **simulation**: generate synthetic data sets with known parameters, run hypothesis tests, and measure the false positive rate. Does it match  $\alpha$ ?



